



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to VMS FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

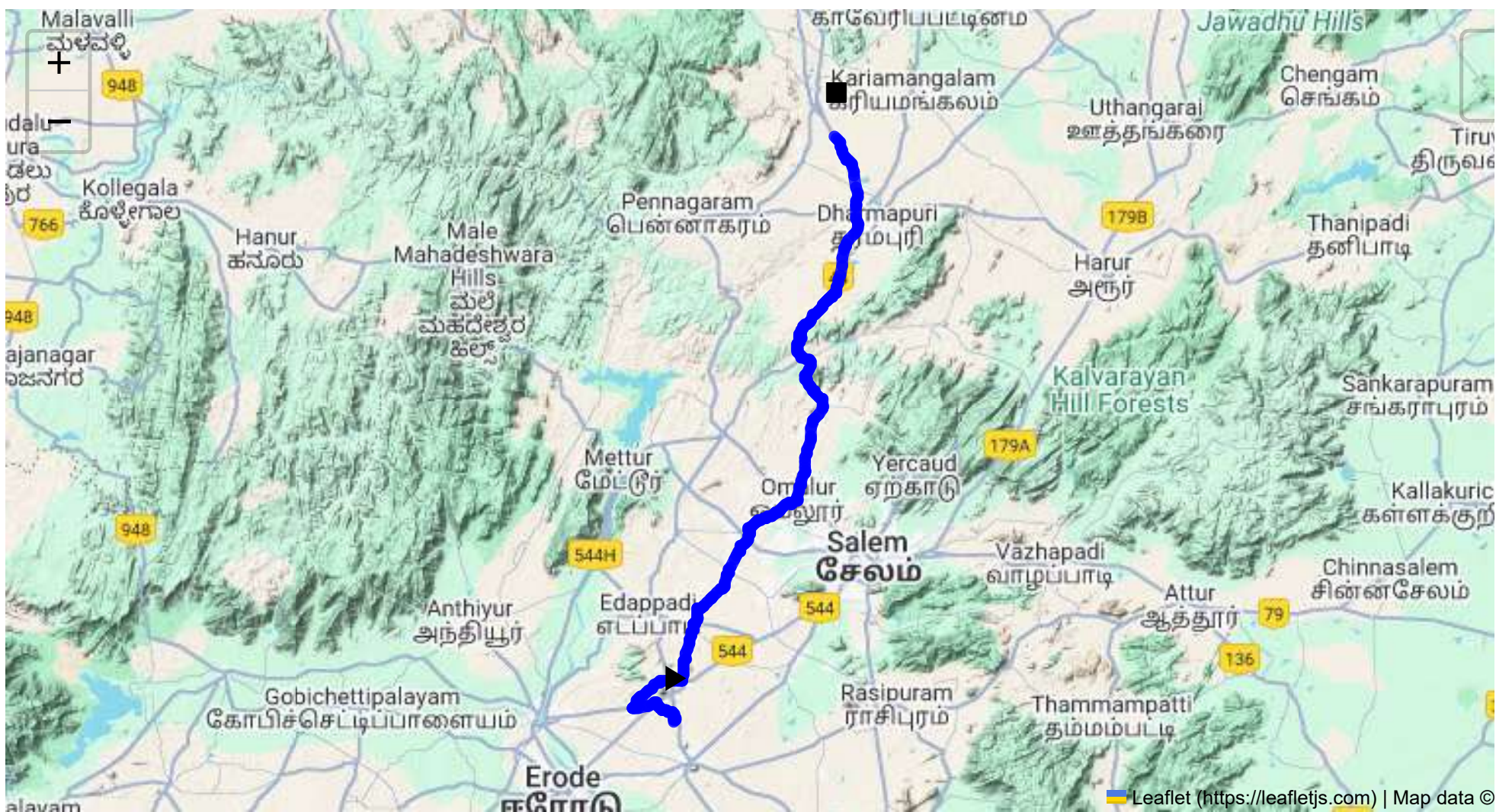
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 116.89 km
Estimated Duration: 2.4 hours
Adjusted Duration (Heavy Vehicle): 3.0 hours
Start: (11.4381, 77.8734)
End: (12.24051, 78.10029)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route covers a significant portion of Tamil Nadu, traversing through various towns and rural areas. Starting from Sangagiri in Tamil Nadu and ending in Somenahalli, the travel path includes key waypoints such as FV2C+2PH Sankari RS - Erode Privu Rd, and several sections along SH-86. The route primarily

follows state highways and major roads, offering a direct path through agricultural and semi-urban regions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** Tamil Nadu generally experiences a tropical climate with hot summers, moderate monsoons, and mild winters. Monsoon season, typically from June to September, can bring heavy rainfall.
- **Potential Hazards:** During monsoons, roads can become slippery, and low-lying areas may face flooding, impacting road conditions and visibility. During peak summer months, high temperatures can cause heat-related vehicle issues and discomfort.

3. Analysis of Traffic Patterns

- **Peak Hours:** Generally, traffic tends to peak in the morning (8:00-10:00 AM) and evening (5:00-8:00 PM) as people commute to and from work.
- **Congestion-prone Areas:** Town centers and market areas in Sankari and Erumaipatti are typically congested. Certain stretches of SH-86 may also see slower traffic, especially near intersections with local roads.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** The primary roads such as SH-86 and Bhavani Main Road are generally maintained but may have patches of wear. Areas in rural sections might have narrower roads and less street lighting.
- **Infrastructure:** Infrastructure is adequate for regular vehicular traffic; however, heavy vehicles might face difficulties on narrower village roads or where road maintenance is pending.

5. Suggestions for Alternative Routes for Emergencies

In emergencies, SH-86 offers several offshoot routes connecting to NH-44, a major national highway that might provide better access to emergency services and faster routes out of congested areas.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires adherence to specific regulations such as obtaining necessary permits and displaying proper signage. Restrictions may apply during certain hours in urban areas to minimize risk and congestion.

7. Overview of Historical Incidents

While no specific incidents may have been documented publicly, local reports suggest common risks include vehicle breakdowns and accidents in congested areas. Hazardous material spills, if any, would likely have stricter historical documentation due to regulatory oversight.

8. Environmental Considerations and Sensitive Areas

- Water Bodies:** Be cautious near river or canal crossings, especially during monsoon seasons when water levels rise.
- Wildlife Areas:** Some stretches may be near farming zones, approaching wildlife cautiously as stray animals can interfere with traffic.

9. Analysis of Communication Coverage

The route enjoys relatively good mobile coverage, but rural patches, especially near agricultural lands, might have spotty signal strength leading to potential communication dead zones.

10. Estimated Emergency Response Times

- Urban Areas:** Response times are typically quicker in and around towns such as Sangagiri and Erumaipatti (within 30 minutes).
- Rural Sections:** More isolated areas may experience longer response times, potentially exceeding 45 minutes to an hour. It is advisable to clarify procedures with local service providers.

12. Overall Summary of Risk Assessment

Overall, the route is moderately safe with careful attention needed during peak traffic hours, poor weather conditions, and in rural segments. Proper planning, compliance with local regulations, and communication readiness can mitigate most risks. It is advisable to frequently monitor weather forecasts, maintain efficient communication equipment, and ensure drivers are well-versed with emergency procedures.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45847, 77.85140	30 KM/Hr
5	Turn	Medium	11.45898, 77.85135	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45571, 77.81702	30 KM/Hr
8	Turn	High	11.45615, 77.81677	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57035, 77.90223	30 KM/Hr
13	Turn	High	11.57041, 77.90221	15 KM/Hr
14	Turn	Medium	11.57649, 77.90610	30 KM/Hr
15	Turn	Medium	11.58279, 77.90658	30 KM/Hr
16	Turn	Medium	11.63833, 77.94818	30 KM/Hr
17	Turn	Medium	11.64171, 77.95184	30 KM/Hr
18	Turn	Medium	11.64243, 77.95198	30 KM/Hr
19	Turn	Medium	11.64721, 77.95118	30 KM/Hr
20	Turn	Medium	11.67870, 77.96846	30 KM/Hr
21	Turn	High	11.70227, 77.97864	15 KM/Hr
22	Turn	Medium	11.71715, 77.99969	30 KM/Hr
23	Turn	Medium	11.71921, 78.01133	30 KM/Hr
24	Turn	Medium	11.73775, 78.03465	30 KM/Hr
25	Turn	High	11.73651, 78.03878	15 KM/Hr
26	Turn	Medium	11.74309, 78.04725	30 KM/Hr
27	Turn	Medium	11.86873, 78.08374	30 KM/Hr
28	Turn	Medium	11.87966, 78.08187	30 KM/Hr
29	Turn	Medium	11.92609, 78.06513	30 KM/Hr
30	Turn	Medium	11.93719, 78.05372	30 KM/Hr
31	Turn	Medium	11.95457, 78.04685	30 KM/Hr
32	Turn	High	11.95828, 78.04983	15 KM/Hr
33	Turn	Medium	11.95885, 78.04988	30 KM/Hr
34	Turn	Medium	11.96044, 78.04949	30 KM/Hr
35	Turn	Medium	11.96335, 78.05141	30 KM/Hr
36	Turn	Medium	11.96408, 78.05131	30 KM/Hr
37	Turn	Medium	11.96511, 78.04966	30 KM/Hr
38	Turn	High	11.96608, 78.04919	15 KM/Hr
39	Turn	Medium	11.96819, 78.05088	30 KM/Hr
40	Turn	Medium	12.12075, 78.13278	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
14	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
15	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
16	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
18	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
19	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
30	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49465, 77.88488



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56096, 77.89931



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57035, 77.90223



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57041, 77.90221



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57649, 77.90610



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58279, 77.90658



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63833, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64171, 77.95184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64243, 77.95198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67870, 77.96846



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70227, 77.97864



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71715, 77.99969



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71921, 78.01133



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73775, 78.03465



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03878



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74309, 78.04725



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86873, 78.08374



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87966, 78.08187



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92609, 78.06513



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93719, 78.05372



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95457, 78.04685



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95828, 78.04983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95885, 78.04988



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96044, 78.04949



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96335, 78.05141



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96408, 78.05131



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96511, 78.04966



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96608, 78.04919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96819, 78.05088



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.12075, 78.13278
